** Project Day 6 **

# Working on Real Project with Python on 'London Housing Dataset'

(A part of Big Data Analysis)

## LONDON HOUSING DATASET

This dataset is primarily centered around the housing market of London. It contains a lot of additional relevant data:

-> Monthly average house prices

-> Yearly number of houses sold

-> Monthly number of crimes committed

The data used here is from year 1995 to 2019 of each different area.

In [345…

```python
import pandas as pd
```

In [347…

```python
df=pd.read_csv('london house dataset.csv')
```

In [349…

```python
df
```

Out[349…

| | date | area | average_price | code | houses_sold | no_of_crimes |
|---|---|---|---|---|---|---|
| 0 | 1/1/1995 | city of london | 91449 | E09000001 | 17.0 | NaN |
| 1 | 2/1/1995 | city of london | 82203 | E09000001 | 7.0 | NaN |
| 2 | 3/1/1995 | city of london | 79121 | E09000001 | 14.0 | NaN |
| 3 | 4/1/1995 | city of london | 77101 | E09000001 | 7.0 | NaN |
| 4 | 5/1/1995 | city of london | 84409 | E09000001 | 10.0 | NaN |
| ... | ... | ... | ... | ... | ... | ... |
| 13544 | 9/1/2019 | england | 249942 | E92000001 | 64605.0 | NaN |
| 13545 | 10/1/2019 | england | 249376 | E92000001 | 68677.0 | NaN |
| 13546 | 11/1/2019 | england | 248515 | E92000001 | 67814.0 | NaN |
| 13547 | 12/1/2019 | england | 250410 | E92000001 | NaN | NaN |
| 13548 | 1/1/2020 | england | 247355 | E92000001 | NaN | NaN |

13549 rows × 6 columns

In [350…

```python
df.count()
```

Out[350…

```
date             13549
area             13549
average_price    13549
code             13549
houses_sold      13455
no_of_crimes      7439
dtype: int64
```

In [351…

```python
df.isnull().sum()
```

Out[351…

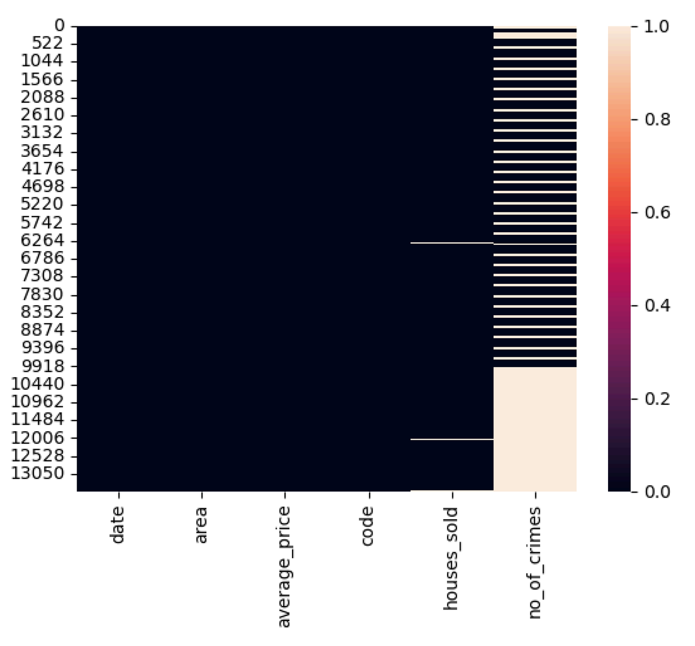
```
date                0
area                0
average_price       0
code                0
houses_sold        94
no_of_crimes     6110
dtype: int64
```

In [355…

```python
import seaborn as sns
import matplotlib.pyplot as plt
```

In [357…

```python
sns.heatmap(df.isnull())
plt.show()
```




## (A) Convert the Datatype pf 'Date' column to Date-Time format.

In [359…

```python
df.dtypes
```

Out[359…

```
date              object
area              object
average_price      int64
code              object
houses_sold      float64
no_of_crimes     float64
dtype: object
```

In [362…

```python
df.date=pd.to_datetime(df.date)
```

In [364…

```python
df.dtypes
```

Out[364…

```
date             datetime64[ns]
area                     object
average_price             int64
code                     object
houses_sold             float64
no_of_crimes            float64
dtype: object
```

## (B.1) Add a new column "year" in the dataframe, which contains year only.

In [278…

```python
df.head(2)
```

Out[278…

| | date | area | average_price | code | houses_sold | no_of_crimes |
|---|---|---|---|---|---|---|
| 0 | 1995-01-01 | city of london | 91449 | E09000001 | 17.0 | NaN |
| 1 | 1995-02-01 | city of london | 82203 | E09000001 | 7.0 | NaN |

In [304…

```python
df['Year']=df.date.dt.year
# df['Month']=df.date.dt.month
```

In [306…

```python
df.head(2)
```

Out[306…

| | date | area | average_price | code | houses_sold | no_of_crimes | Year |
|---|---|---|---|---|---|---|---|
| 0 | 1995-01-01 | city of london | 91449 | E09000001 | 17.0 | NaN | 1995 |
| 1 | 1995-02-01 | city of london | 82203 | E09000001 | 7.0 | NaN | 1995 |

## (B.2) Add a new column "month" as 2nd column in the dataframe, which contains month only.

In [285…

```python
df.insert( 2, 'month', df.date.dt.month)
```

In [287…

```python
df.head(3)
```

Out[287…

| | date | area | month | average_price | code | houses_sold | no_of_crimes | Year |
|---|---|---|---|---|---|---|---|---|
| 0 | 1995-01-01 | city of london | 1 | 91449 | E09000001 | 17.0 | NaN | 1995 |
| 1 | 1995-02-01 | city of london | 2 | 82203 | E09000001 | 7.0 | NaN | 1995 |
| 2 | 1995-03-01 | city of london | 3 | 79121 | E09000001 | 14.0 | NaN | 1995 |

## (c) Remove the columns 'year' and 'month' from the dataframe.

In [290…

```python
# df.drop(['month','Year'],axis=1,inplace=True)
```

In [292…

```python
# df.head(2)
```

Out[292…

| | date | area | average_price | code | houses_sold | no_of_crimes |
|---|---|---|---|---|---|---|
| 0 | 1995-01-01 | city of london | 91449 | E09000001 | 17.0 | NaN |
| 1 | 1995-02-01 | city of london | 82203 | E09000001 | 7.0 | NaN |

## (D) Show all the records where 'No. of Crimes' is 0. and, how many such records are there ?

In [295…

```python
df[df.no_of_crimes==0]
```

Out[295…

| | date | area | average_price | code | houses_sold | no_of_crimes |
|---|---|---|---|---|---|---|
| 72 | 2001-01-01 | city of london | 284262 | E09000001 | 24.0 | 0.0 |
| 73 | 2001-02-01 | city of london | 198137 | E09000001 | 37.0 | 0.0 |
| 74 | 2001-03-01 | city of london | 189033 | E09000001 | 44.0 | 0.0 |
| 75 | 2001-04-01 | city of london | 205494 | E09000001 | 38.0 | 0.0 |
| 76 | 2001-05-01 | city of london | 223459 | E09000001 | 30.0 | 0.0 |
| ... | ... | ... | ... | ... | ... | ... |
| 178 | 2009-11-01 | city of london | 397909 | E09000001 | 11.0 | 0.0 |
| 179 | 2009-12-01 | city of london | 411955 | E09000001 | 16.0 | 0.0 |
| 180 | 2010-01-01 | city of london | 464436 | E09000001 | 20.0 | 0.0 |
| 181 | 2010-02-01 | city of london | 490525 | E09000001 | 9.0 | 0.0 |
| 182 | 2010-03-01 | city of london | 498241 | E09000001 | 15.0 | 0.0 |

104 rows × 6 columns

In [297…

```python
len(df[df.no_of_crimes==0])
```

Out[297… 104

## (E) What is the maximum & minimum 'Average_price' per year in England ?

In [308…

```python
df1=df[df.area =='england']
```

In [310…

```python
df1.groupby('Year').average_price.max().sum()
```

Out[310… 4192410

In [312…

```python
df1.groupby('Year').average_price.min().sum()
```

Out[312… 3909794

In [314…

```python
df1.groupby('Year').average_price.mean().sum()
```

Out[314… 4066218.9166666665

## (F) What is the Maximum & Minimum No. of Crimes recorded per area ?

In [317…

```python
df.groupby('area').no_of_crimes.max()
```

Out[317…

```
area
barking and dagenham      2049.0
barnet                    2893.0
bexley                    1914.0
brent                     2937.0
bromley                   2637.0
camden                    4558.0
city of london              10.0
croydon                   3263.0
ealing                    3401.0
east midlands                NaN
east of england              NaN
enfield                   2798.0
england                      NaN
greenwich                 2853.0
hackney                   3466.0
hammersmith and fulham    2645.0
haringey                  3199.0
harrow                    1763.0
havering                  1956.0
hillingdon                2819.0
hounslow                  2817.0
inner london                 NaN
islington                 3384.0
kensington and chelsea    2778.0
kingston upon thames      1379.0
lambeth                   4701.0
lewisham                  2813.0
london                       NaN
merton                    1623.0
newham                    3668.0
north east                   NaN
north west                   NaN
outer london                 NaN
redbridge                 2560.0
richmond upon thames      1551.0
south east                   NaN
south west                   NaN
southwark                 3821.0
sutton                    1425.0
tower hamlets             3316.0
waltham forest            2941.0
wandsworth                3051.0
west midlands                NaN
westminster               7461.0
yorks and the humber         NaN
Name: no_of_crimes, dtype: float64
```

In [319…

```python
df.groupby('area').no_of_crimes.max().sort_values(ascending=True)
```

Out[319…

```
area
city of london              10.0
kingston upon thames      1379.0
sutton                    1425.0
richmond upon thames      1551.0
merton                    1623.0
harrow                    1763.0
bexley                    1914.0
havering                  1956.0
barking and dagenham      2049.0
redbridge                 2560.0
bromley                   2637.0
hammersmith and fulham    2645.0
kensington and chelsea    2778.0
enfield                   2798.0
lewisham                  2813.0
hounslow                  2817.0
hillingdon                2819.0
greenwich                 2853.0
barnet                    2893.0
brent                     2937.0
waltham forest            2941.0
wandsworth                3051.0
haringey                  3199.0
croydon                   3263.0
tower hamlets             3316.0
islington                 3384.0
ealing                    3401.0
hackney                   3466.0
newham                    3668.0
southwark                 3821.0
camden                    4558.0
lambeth                   4701.0
westminster               7461.0
east midlands                NaN
east of england              NaN
england                      NaN
inner london                 NaN
london                       NaN
north east                   NaN
north west                   NaN
outer london                 NaN
south east                   NaN
south west                   NaN
west midlands                NaN
yorks and the humber         NaN
Name: no_of_crimes, dtype: float64
```

## (G) Show the total count of records of each area,where average price is less than 100000.

In [327…

```python
df[df.average_price < 100000].area.value_counts()
```

Out[327…

```
area
north east              112
north west              111
yorks and the humber    110
east midlands            96
west midlands            94
england                  87
barking and dagenham     85
south west               78
east of england          76
newham                   72
bexley                   64
waltham forest           64
lewisham                 62
havering                 60
south east               59
greenwich                59
croydon                  57
enfield                  54
sutton                   54
hackney                  53
redbridge                52
southwark                48
tower hamlets            47
outer london             46
hillingdon               44
lambeth                  41
hounslow                 41
brent                    40
london                   39
merton                   35
haringey                 33
bromley                  33
inner london             31
ealing                   31
kingston upon thames     30
harrow                   30
wandsworth               26
barnet                   25
islington                19
city of london           11
Name: count, dtype: int64
```

End - Project

In [ ]: